

Customer Churn Analysis

Task 2 – Data Quality and Exploratory Analysis

Objective: Create a reproducible, analysis-ready customer churn dataset and document data-quality decisions and initial patterns.

1. Dataset Overview

Item	Result
Records	15 customers
Columns	11 fields
Missing values	None visible in the provided dataset
Duplicate Customer IDs	None identified
Churned customers	7
Non-churned customers	8
Churn rate	46.67%
Retention rate	53.33%

2. Data Quality Assessment

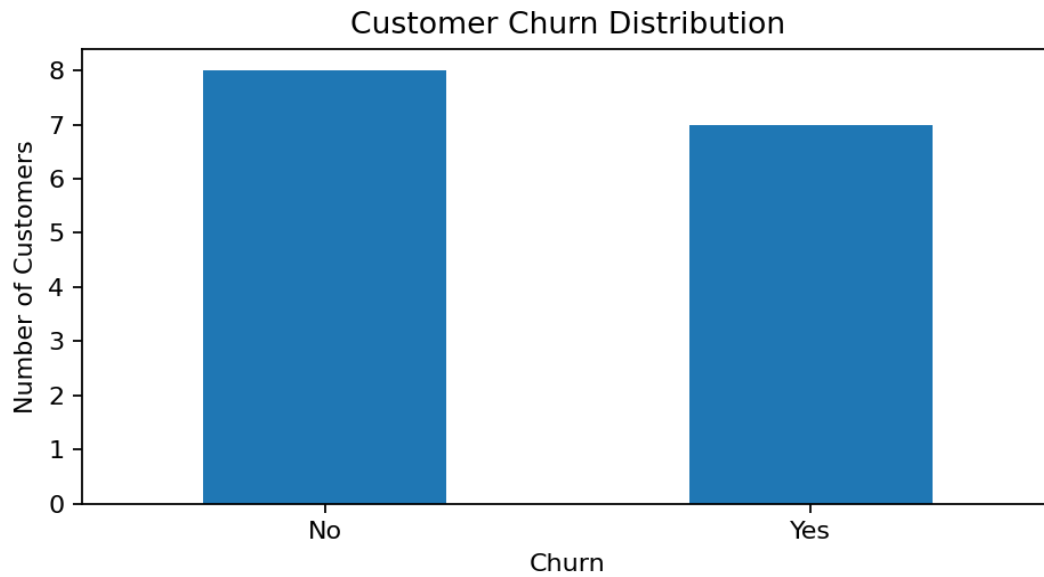
The displayed customer churn data was checked for completeness, uniqueness, valid ranges, and consistency. No missing cells or duplicate Customer IDs are visible in the supplied 15-record dataset. Numerical fields are suitable for statistical analysis, while categorical fields use consistent labels. No value was removed because of an obvious invalid entry.

Check	Finding	Decision
Missing values	None visible	No imputation required
Duplicate records	None identified	No deletion required
Customer IDs	15 unique IDs	Retain
Age	23–60 years	No obvious invalid value
Tenure	2–48 months	Retain
Monthly charge	■49.99–■149.99	Retain
Support tickets	0–6	Retain
Churn	Yes / No only	Consistent

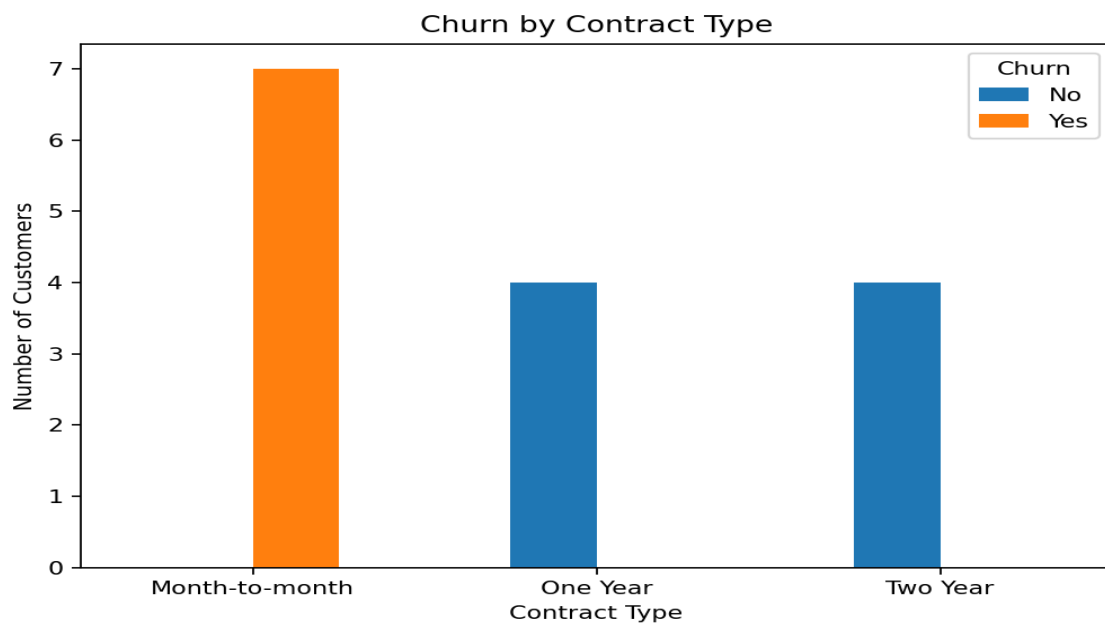
3. Summary Statistics

Variable	Mean	Median	Minimum	Maximum
Age	39.07	38	23	60
Tenure (months)	19.47	18	2	48
Monthly Charge	■84.66	■79.99	■49.99	■149.99
Total Charges	■2141.81	■1119.86	■99.98	■7199.52
Support Tickets	2.53	2	0	6

4. Exploratory Data Analysis



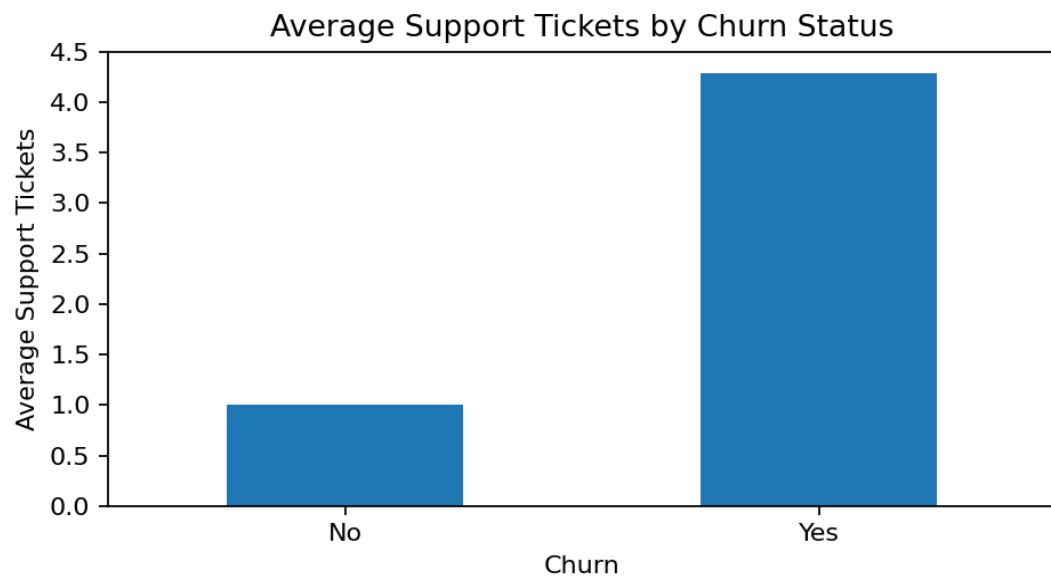
The sample contains 7 churned customers and 8 customers who did not churn, producing a churn rate of 46.67%.



Contract pattern: all 7 month-to-month customers in this sample churned, while none of the 4 one-year or 4 two-year customers churned. This is a strong pattern in this small sample, but it should not be treated as proof of causation.

5. Subscription and Support Analysis

Subscription	Not Churned	Churned
Basic	2	5
Pro	2	2
Enterprise	4	0



6. Churned vs Non-Churned Comparison

Metric	Churned (Yes)	Not Churned (No)
Average Age	30.43	46.62
Average Tenure	8.43 months	29.12 months
Average Monthly Charge	■58.56	■107.49
Average Support Tickets	4.29	1.00

7. Key Findings

1. The dataset contains 15 customers with no visible missing values or duplicate Customer IDs.
2. The observed churn rate is 46.67% (7 of 15 customers).
3. Month-to-month contracts show the highest churn in this sample: all 7 month-to-month customers churned.
4. One-year and two-year contract customers show no churn in the displayed sample.
5. Basic subscription customers account for 5 of the 7 churned customers.
6. Churned customers have a much shorter average tenure (8.43 months) than non-churned customers (29.13 months).
7. Churned customers average 4.29 support tickets versus 1.00 for non-churned customers.

8. Conclusion

The cleaned dataset is suitable for initial exploratory analysis. The strongest observed patterns are associated with contract type, customer tenure, subscription level, and support-ticket volume. Because the dataset contains only 15 records, these findings should be considered exploratory rather than statistically conclusive. A larger customer sample would be required before making business-wide retention decisions.

Prepared for: Task 2 – Data Quality and Exploratory Analysis

Source: Customer churn dataset shown in the supplied Excel screenshot.